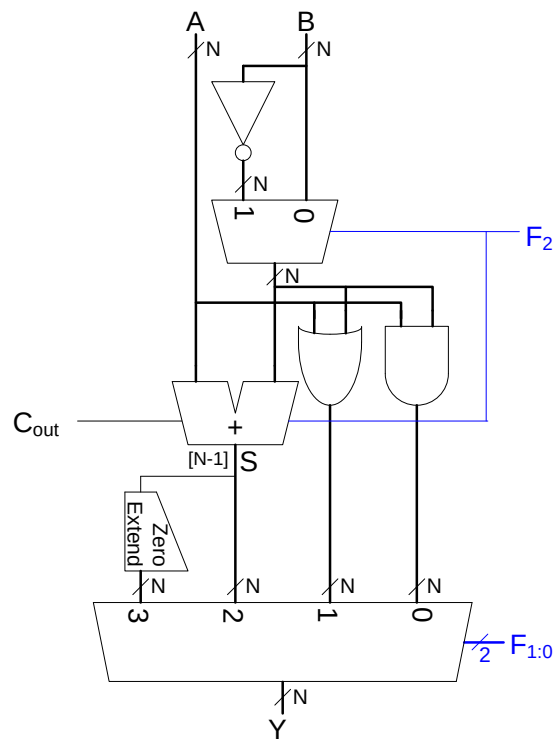
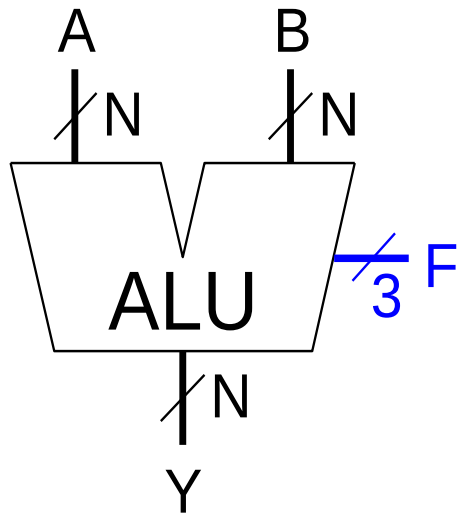


Lab3: Arithmetic and Logic Unit (ALU) and Register File

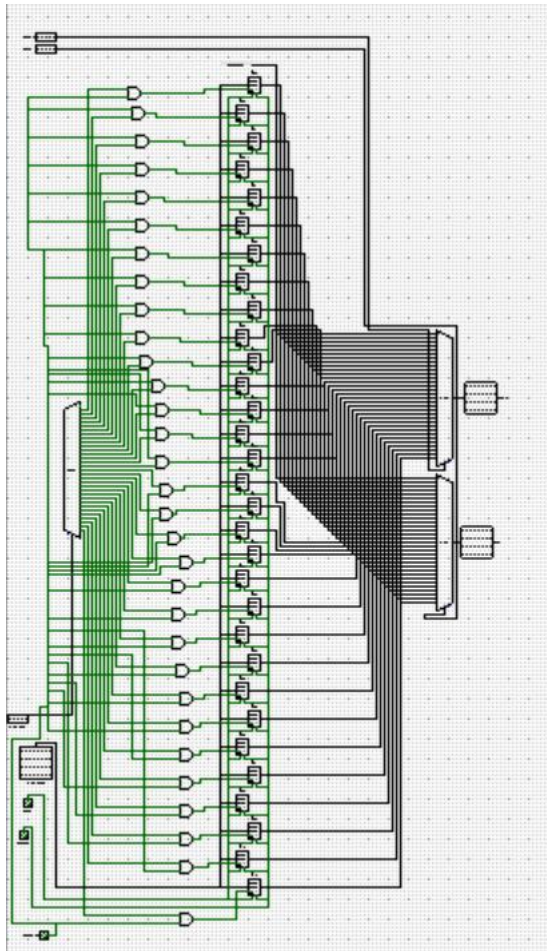
T1: Design and Test the ALU shown in Figure.





$F_{2:0}$	Function
000	$A \& B$
001	$A B$
010	$A + B$
011	not used
100	$A \& \sim B$
101	$A \sim B$
110	$A - B$
111	SLT

T2: Design 32x32 register file with one write port and two read port.



T3: Combine the register file and ALU and test the ALU by taking the operands from the ALU. For example Test ALU by reading two register and write

Submission

Show your simulation to TAs. T1-T3 showing design/details should be complete in your record. Submit your single library (Lab3_roll_no_FirstName.circ file) to:

<https://u.pcloud.com/#/puplink?code=U4WXZqo3oNwaRlGYqRDfWt1BpdkEXuuAk>

(Use **your Roll number** instead of Name for **submission**)